Matthew Emmanuel T. Labrador

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EDUCATION

University of the Philippines Manila *Manila, PH*

BS Computer Science, University Scholar | Expected May 2029

Current Cumulative GWA: 1.0375 (Summa Cum Laude standing) • DOST Undergraduate Scholar

PAREF Southridge School *Manila, PH*

Senior High School, Class Valedictorian and Gold Medalist | May 2025

Mercury Drug 2025 Gawad Talindaw Award for Excellence in Science

EXPERIENCE AND PROJECTS

NVIDIA Philippines | Campus Ambassador *2026–Present*

PC Express | Intern *2026–Present*

- Across these roles, develop local AI applications with NVIDIA RTX 5090 GPUs and deliver talks and training on using and deploying models on personal hardware.

Tumbang Preso (TUMP) | Gameplay & Systems Engineer *2026*

Godot 4, GDScript, Blender, ENet UDP Authoritative Netcode, Dedicated Linux Servers

- Won Grand Champion with Team BH Studios at Gear Up NCR Esports Game Dev Challenge; selected as NCR Representative to National Finals (backed by DOST & Level Up).
- Engineered core multiplayer systems from empty project to shipped build in 5 days: gameplay mechanics, 3D asset pipeline, sound design, bot AI, and netcode.
- Engineered authoritative-host netcode over ENet with distance-based collision validation after profiling and discovering 16 of 36 built-in engine physics overlap callbacks silently dropped under network jitter.
- Deployed dedicated online lobbies on a Linux VPS with UDP discovery and join codes, reaching players across different networks and CGNAT.

eGovMed & PGH OCRA Mobile | Full-Stack Lead & Systems Architect *2026*

React 18/19, Vite, Capacitor (Android/iOS), Node.js, Express, TypeScript, Redis, Server-Sent Events, 8 eGovPH APIs

- Grand Champion at eGov Hackathon PH 2026 (₱100,000 prize), with Philippine General Hospital as the pilot target.
- Engineered cross-platform mobile app & PWA featuring bilingual AI triage with deterministic safety floors (100% emergency recall, 0% false alarms).
- Architected real-time queue stream over SSE with dynamic rolling-median ETAs and cross-instance Redis fan-out (load-tested to 6,000 concurrent streams at <600ms latency); integrated 8 national eGovPH APIs with atomic Redis compare-and-set liveness locks.
- Authored 127 automated security & safety tests, envelope encryption under Data Privacy Act, and VAPID web push notifications.

Knee MRI Reader | Deep Learning & Computer Vision Engineer *2026*

Python, PyTorch, DINOv2 (ViT), Cross-View Attention, ONNX Runtime, Next.js 16, React 19, TypeScript

- Trained an end-to-end diagnostic model scoring 12 knee pathologies across 4,407 DICOM MRI studies, achieving 0.843 Out-of-Fold Macro ROC-AUC (0.96 Medial Osteoarthritis, 0.95 Effusion/Baker's Cyst, 0.90 Fracture, 0.88 ACL).
- Designed multi-view cross-attention architecture: mapped varying patient series into 6 canonical anatomical planes using directional cosine vectors, extracting frozen DINOv2 patch features with 12 specialized diagnostic attention heads.
- Built a 3-LLM weak supervision consensus engine on 4,349 multi-lingual free-text reports, boosting label accuracy from 0.87 to 0.89.
- Deployed real-time workstation: exported a 20-model ensemble to sub-200ms ONNX models with interactive client-side DICOM slice parsing.

Diabetic Complication Swarm (GlycoSwarm AI) | Multi-Agent Systems Lead *2026*

Python, LangGraph, FastAPI, Ollama, Gemma 4 26B, Fireworks GLM 5.2, AMD MI300X GPU, Next.js

- Lead developer of international team at AMD Hackathon ACT II building a LangGraph StateGraph of 4 parallel specialist agents (Renal, Neuropathy, Retinal, Cardiovascular) that dynamically generate and execute Python scoring scripts against real NHANES lab data.
- Served live LLM inference on AMD MI300X accelerators via Ollama with automated failover routing to Fireworks GLM.

Khanpanion | Team Lead & Developer *2026*

Next.js, React, TypeScript, Khan Academy Resources, Rule-Based Remediation

- Grand Champion, Khan Academy Education Impact Challenge 2026 (Enactus Philippines).
- Developed a Next.js study platform with typed prerequisite routing, generated practice, resumable mock exams and device-local learning state, linked to targeted Khan Academy resources.

Heart Disease Prediction Model | Machine Learning Engineer *2025*

Python, scikit-learn, pandas, Streamlit, Google Colab

- Trained a supervised machine learning classification pipeline in Python to predict cardiovascular disease risk from patient clinical features end-to-end.
- Performed data preprocessing, missing value imputation, and categorical feature engineering using pandas and scikit-learn.

TECHNICAL SKILLS

Languages:	Python, SQL, C, C++17, TypeScript, JavaScript, GDScript, HTML
ML & Data:	Feature Engineering, Evaluation Metrics (ROC-AUC, Precision, Recall, F1), pandas, Prompt Engineering, Model Deployment
Agentic AI & Orch:	PyTorch, DINOv2 (ViT), ONNX Runtime, LangChain, LangGraph, Multi-Agent Systems, FastAPI, Ollama, RAG, SHAP
Web & Mobile:	Next.js 16, React 19/18, Node.js, Express, Vite, Tailwind CSS, Capacitor (Android/iOS), PWA, Redis, SSE, VAPID Web Push
Systems & Tools:	Godot 4, ENet Authoritative Netcode, Raylib, CMake, WebAssembly, Blender, Linux VPS, Git/CI-CD (GitHub Actions)
Certifications:	AMD Multiagent Systems Deployment, CS50 Intro to CS, AI Prompt Engineering, Python, C

LEADERSHIP AND ACTIVITIES

Competitive Debater *2022–2025*

Quarterfinalist at Ateneo Peace Debate; competed at NASH DC (DLSU), PSDC (Ateneo), UP Diliman Debates, XSDC, and ASDC.

Math Team Captain — PAREF Southridge School *2022–2024*

Led tutoring sessions and mentored student peers for local and regional inter-school mathematics competitions.

Google Developer Group & UP Socomsci *2025–Present*

Coordinated cross-functional student developer teams on operations and event setup for tech conferences and workshops.

AWARDS

Grand Champion, Khan Academy Education Impact Challenge (2026) • Grand Champion, eGov Hackathon PH (2026, ₱100,000) • Grand Champion, Gear Up NCR (2026, NCR Representative to Nationals) • DOST Undergraduate Scholar (2025–Present) • Top 5 Finalist, Olymphysics NCR (2025) • 5th Place, Philippine Statistics Quiz NCR (2025)